# Product Measure for Fubini and Tonelli Theorems: Examples and Proofs

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## 1 Introduction

The purpose of this project is to explain product measure and its connection with multivariable integration. The main source is Chapter 5 of Sheldon Axler's *Measure, Integration & Real Analysis*, where product $\sigma$-algebras, product measures, Tonelli's theorem, and Fubini's theorem are developed. This project is written for readers who already know the basic language of measure theory and integration, but have not yet studied the product measure chapter carefully.

The goal is to make double integrals less mysterious: product measure gives a precise meaning to measuring subsets of $X \times Y$, and Tonelli's and Fubini's theorems explain when a product integral can be computed by iterated integrals.

## 2 Definitions and Examples

**Definition 1** (Measurable space and $\sigma$-algebra)**.** A measurable space is a pair $(X, \mathcal{S})$ where $X$ is a set and $\mathcal{S}$ is a $\sigma$-algebra on $X$. This means $\mathcal{S}$ is a collection of subsets of $X$ such that $X \in \mathcal{S}$, $X \setminus A \in \mathcal{S}$ whenever $A \in \mathcal{S}$, and $\bigcup_{n=1}^{\infty} A_n \in \mathcal{S}$ whenever each $A_n \in \mathcal{S}$.

**Example 1.** If $X = \{a, b, c\}$, then $\mathcal{S} = \{\varnothing, X, \{a\}, \{b, c\}\}$ is a $\sigma$-algebra. It contains $X$ and $\varnothing$, it is closed under complements, and any countable union eventually repeats one of these four sets. This example is small, but it shows the basic idea: a $\sigma$-algebra tells us which subsets are allowed to be measured.

**Definition 2** (Measurable rectangle and product $\sigma$-algebra)**.** Let $(X, \mathcal{S})$ and $(Y, \mathcal{T})$ be measurable spaces. A measurable rectangle in $X \times Y$ is a set of the form $A \times B$, where $A \in \mathcal{S}$ and $B \in \mathcal{T}$. The product $\sigma$-algebra $\mathcal{S} \otimes \mathcal{T}$ is the smallest $\sigma$-algebra on $X \times Y$ that contains all measurable rectangles.

**Example 2.** Let $X = Y = \mathbb{R}$ and let $\mathcal{S} = \mathcal{T} = \mathcal{B}(\mathbb{R})$, the Borel $\sigma$-algebra. Then $[0, 1] \times [2, 5]$ is a measurable rectangle in $\mathcal{B}(\mathbb{R}) \otimes \mathcal{B}(\mathbb{R})$. More complicated planar sets, such as triangles and disks, are not single rectangles, but they are still in the product $\sigma$-algebra because they can be built from Borel sets by countable set operations.

**Definition 3** (Sections)**.** If $E \subseteq X \times Y$, then for $x \in X$ and $y \in Y$ we define

$$E_x = \{y \in Y : (x, y) \in E\}, \qquad E^y = \{x \in X : (x, y) \in E\}.$$

These are called the vertical and horizontal sections of $E$. If $f : X \times Y \to [-\infty, \infty]$, then we define the section functions

$$f_x(y) = f(x, y), \qquad f^y(x) = f(x, y).$$

**Example 3.** Let $E = [0,1] \times [2,5]$. If $x \in [0,1]$, then $E_x = [2,5]$; if $x \notin [0,1]$, then $E_x = \varnothing$. Similarly, if $y \in [2,5]$, then $E^y = [0,1]$, and otherwise $E^y = \varnothing$. This example explains why sections are useful: they turn a two-variable set into one-variable sets.

**Definition 4** (Product measure)**.** Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be $\sigma$-finite measure spaces. The product measure $\mu \times \nu$ is the measure on $\mathcal{S} \otimes \mathcal{T}$ satisfying

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$$

for every $A \in \mathcal{S}$ and $B \in \mathcal{T}$.

**Example 4.** For Lebesgue measure $\lambda$ on $\mathbb{R}$, the product measure $\lambda \times \lambda$ is the usual two-dimensional Lebesgue measure on $\mathbb{R}^2$. Thus

$$(\lambda \times \lambda)([0,1] \times [2,5]) = \lambda([0,1])\lambda([2,5]) = 1 \cdot 3 = 3.$$

This is exactly the area of the rectangle.

**Definition 5** (Integral of a simple function)**.** A nonnegative simple function has the form $s = \sum_{k=1}^{n} c_k \mathbf{1}_{E_k}$, where $c_k \geq 0$ and the sets $E_k$ are measurable. Its integral is

$$\int s \, d\mu = \sum_{k=1}^{n} c_k \mu(E_k).$$

For instance, if $s = 3\mathbf{1}_{[0,1]\times[0,2]}$ on $\mathbb{R}^2$, then

$$\int_{\mathbb{R}^2} s \, d(\lambda \times \lambda) = 3(\lambda \times \lambda)([0,1] \times [0,2]) = 3 \cdot 2 = 6.$$

## 3 The Main Theorems

The reason product measure is useful is that it lets us compare one integral over $X \times Y$ with two iterated integrals. The two main results are Tonelli's theorem and Fubini's theorem. Tonelli's theorem is for nonnegative functions. Fubini's theorem is for integrable functions.

**Theorem 1** (Tonelli's theorem)**.** *Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be $\sigma$-finite measure spaces. If $f : X \times Y \to [0, \infty]$ is $\mathcal{S} \otimes \mathcal{T}$-measurable, then the functions*

$$x \mapsto \int_Y f(x,y) \, d\nu(y), \qquad y \mapsto \int_X f(x,y) \, d\mu(x)$$

*are measurable, and*

$$\int_{X \times Y} f \, d(\mu \times \nu) = \int_X \int_Y f(x,y) \, d\nu(y) \, d\mu(x) = \int_Y \int_X f(x,y) \, d\mu(x) \, d\nu(y).$$

**Theorem 2** (Fubini's theorem)**.** *Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be $\sigma$-finite measure spaces. Suppose $f : X \times Y \to [-\infty, \infty]$ is $\mathcal{S} \otimes \mathcal{T}$-measurable and*

$$\int_{X \times Y} |f| \, d(\mu \times \nu) < \infty.$$

*Then $f_x \in L^1(\nu)$ for almost every $x \in X$, $f^y \in L^1(\mu)$ for almost every $y \in Y$, and*

$$\int_{X \times Y} f \, d(\mu \times \nu) = \int_X \int_Y f(x,y) \, d\nu(y) \, d\mu(x) = \int_Y \int_X f(x,y) \, d\mu(x) \, d\nu(y).$$

The difference between the two theorems is important. Tonelli does not require the integral to be finite, but it requires $f \geq 0$. Fubini allows positive and negative values, but it requires absolute integrability. In practice, one often uses Tonelli on $|f|$ first; if the integral of $|f|$ is finite, then Fubini can be applied to $f$.

## 4 Annotated Proof of Tonelli's Theorem

Tonelli's theorem is the central proof for this theme, because it is the result that first makes iterated integration work in the nonnegative case. The proof follows a natural measure-theory strategy: first understand indicator functions of rectangles, then extend from rectangles to all measurable sets, then extend from indicator functions to simple functions, and finally use monotone convergence to reach every nonnegative measurable function. This is not just a technical trick; it is the reason nonnegative functions are easier than signed functions.

Start with the most concrete case. Let $E = A \times B$, where $A \in \mathcal{S}$ and $B \in \mathcal{T}$. Then $\mathbf{1}_E(x, y) = \mathbf{1}_A(x)\mathbf{1}_B(y)$. If $x \in A$, the vertical section of $E$ is $B$, so $\int_Y \mathbf{1}_E(x, y)\, d\nu(y) = \nu(B)$. If $x \notin A$, the vertical section is empty, so the same integral is 0. Therefore the inner integral is the one-variable function $\nu(B)\mathbf{1}_A(x)$, and

$$\int_X \int_Y \mathbf{1}_E(x, y)\, d\nu(y)\, d\mu(x) = \int_X \nu(B)\mathbf{1}_A(x)\, d\mu(x) = \mu(A)\nu(B).$$

But by the definition of product measure, $\mu(A)\nu(B) = (\mu \times \nu)(A \times B)$. The same argument with $x$ and $y$ reversed gives the other order of integration. So Tonelli's formula is completely clear for rectangles: it says that measuring a rectangle directly gives the same answer as measuring its slices and then adding the slice sizes.

The next issue is that a general measurable set in $X \times Y$ does not have to be a rectangle. To move beyond rectangles, define $\mathcal{M}$ to be the class of all sets $E \in \mathcal{S} \otimes \mathcal{T}$ for which the section functions are measurable and the equality

$$\int_{X \times Y} \mathbf{1}_E\, d(\mu \times \nu) = \int_X \int_Y \mathbf{1}_E(x, y)\, d\nu(y)\, d\mu(x) = \int_Y \int_X \mathbf{1}_E(x, y)\, d\mu(x)\, d\nu(y)$$

holds. The rectangle calculation shows that every measurable rectangle belongs to $\mathcal{M}$. Finite unions of rectangles also belong to $\mathcal{M}$ because integrals are additive over disjoint pieces. Now suppose $E_n \uparrow E$ and each $E_n$ already belongs to $\mathcal{M}$. Then $\mathbf{1}_{E_n} \uparrow \mathbf{1}_E$, so the Monotone Convergence Theorem lets us pass the equality of integrals from $\mathbf{1}_{E_n}$ to $\mathbf{1}_E$. This is the key reason monotone convergence appears in the proof: increasing unions of sets become increasing limits of indicator functions. A monotone class argument then says that, since rectangles generate $\mathcal{S} \otimes \mathcal{T}$, the equality holds for every $E \in \mathcal{S} \otimes \mathcal{T}$.

Once the theorem is known for indicator functions, the simple-function case follows by linearity. If $s = \sum_{k=1}^{n} c_k \mathbf{1}_{E_k}$ with $c_k \geq 0$ and $E_k \in \mathcal{S} \otimes \mathcal{T}$, then each $\mathbf{1}_{E_k}$ satisfies the Tonelli equality, so adding the equalities with weights $c_k$ gives

$$\int_{X \times Y} s\, d(\mu \times \nu) = \int_X \int_Y s(x, y)\, d\nu(y)\, d\mu(x) = \int_Y \int_X s(x, y)\, d\mu(x)\, d\nu(y).$$

This is the bridge from sets to functions: a simple function is just a finite weighted combination of indicator functions.

Finally, let $f : X \times Y \to [0, \infty]$ be measurable. There exists an increasing sequence of nonnegative simple functions $s_n$ such that $s_n \uparrow f$ pointwise. For every $n$, Tonelli's equality holds for $s_n$. Since $s_n$ increases to $f$, the Monotone Convergence Theorem can be applied on the product space and also inside each iterated integral. Taking the limit as $n \to \infty$ gives exactly

$$\int_{X\times Y} f\, d(\mu \times \nu) = \int_X \int_Y f(x, y)\, d\nu(y)\, d\mu(x) = \int_Y \int_X f(x, y)\, d\mu(x)\, d\nu(y).$$

This proves Tonelli's theorem. The proof also explains the intuition: nonnegative measurable functions can be built from below by simple functions, and for simple functions the theorem is just the rectangle formula plus additivity.

## 5 Example: Fubini Can Fail Without Its Hypotheses

Fubini's theorem is powerful, but it is not automatic. The following example shows why the absolute integrability assumption matters. Define

$$f(x, y) = \begin{cases} \dfrac{x^2 - y^2}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases}$$

on $[0, 1] \times [0, 1]$. For fixed $x > 0$,

$$\int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\, dy = \left[ \frac{y}{x^2 + y^2} \right]_0^1 = \frac{1}{x^2 + 1}.$$

Therefore

$$\int_0^1 \int_0^1 f(x, y)\, dy\, dx = \int_0^1 \frac{1}{x^2 + 1}\, dx = \frac{\pi}{4}.$$

But for fixed $y > 0$,

$$\int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\, dx = \left[ -\frac{x}{x^2 + y^2} \right]_0^1 = -\frac{1}{1 + y^2}.$$

Therefore

$$\int_0^1 \int_0^1 f(x, y)\, dx\, dy = -\frac{\pi}{4}.$$

The two iterated integrals are different. This does not contradict Fubini's theorem, because $f$ is not nonnegative and is not absolutely integrable near the origin. In polar coordinates, $|f|$ behaves like $|\cos(2\theta)|/r^2$, and after the Jacobian factor $r$ is included, the radial behavior is like $1/r$, which is not integrable near 0.

## 6 Two Examples of Product Integration

**Example 5** (A very simple rectangle)**.** Let $f = \mathbf{1}_{[0,1]\times[2,5]}$ on $\mathbb{R}^2$. For a fixed $x$, the vertical section has length 3 if $x \in [0, 1]$ and length 0 otherwise. Hence

$$\int_{\mathbb{R}^2} f\, d(\lambda \times \lambda) = \int_0^1 \int_2^5 1\, dy\, dx = \int_0^1 3\, dx = 3.$$

This is the product-measure version of the rectangle formula: area equals length times width.

**Example 6** (A slightly more interesting triangular region)**.** Let

$$T = \{(x, y) \in \mathbb{R}^2 : 0 \leq y \leq x \leq 1\}.$$

Using vertical sections, for each $x \in [0, 1]$ the possible $y$ values are $0 \leq y \leq x$, so

$$(\lambda \times \lambda)(T) = \int_0^1 \int_0^x 1 \, dy \, dx = \int_0^1 x \, dx = \frac{1}{2}.$$

Using horizontal sections, for each $y \in [0, 1]$ the possible $x$ values are $y \leq x \leq 1$, so

$$(\lambda \times \lambda)(T) = \int_0^1 \int_y^1 1 \, dx \, dy = \int_0^1 (1 - y) \, dy = \frac{1}{2}.$$

This example is still simple, but it shows the real use of sections: changing the order of integration changes the description of the slices, not the set being measured.

## Conclusion

Product measure gives the correct language for double integration. Tonelli proves the nonnegative case, while Fubini needs absolute integrability. The main lesson is that changing the order of integration is justified by hypotheses, not just by notation.

## References


Sheldon Axler, *Measure, Integration & Real Analysis*, Chapter 5: Product Measures. Available at https://measure.axler.net/MIRA.pdf.